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Evaluation of methods to estimate nocturnal bird migration activity: A comparison of radar and nocturnal flight call monitoring in the American West

Dylan M. Osterhaus,[1,2,*] Benjamin M. Van Doren,[3] Kyle G. Horton,[4] Fitsum Abadi,[2] Timothy F. Wright,[1] and Martha J. Desmond[2]

[1]Department of Biology, New Mexico State University, Las Cruces, New Mexico, USA
[2]Department of Fish, Wildlife, and Conservation Ecology, New Mexico State University, Las Cruces, New Mexico, USA
[3]Department of Natural Resources and Environmental Sciences, University of Illinois at Urbana-Champaign, Urbana, Illinois, USA
[4]Department of Fish, Wildlife, and Conservation Biology, Colorado State University, Fort Collins, Colorado, USA
[*]Corresponding author: dylano@nmsu.edu

ABSTRACT

Bird migration involves the movements of billions of individuals but is difficult to study because it occurs primarily at night. We sought to improve our understanding of the methods available to study migration, particularly in understudied regions of western North America. We evaluated 2 methods: weather radar and nocturnal flight call monitoring. We analyzed variability in estimates of migration activity from each method, how estimates relate, and identified factors associated with variation in this relationship. We collected radar and flight call data from southern New Mexico in western North America during the fall migration of 2021 and 2022. Similar studies have occurred in eastern North America, but it is unknown if regional variability alters the relationship between estimates from each method. We found that estimates were positively related across a season, but relationships were variable among nights. Also, we discovered that the strength of the association between methods varied across sites, indicating that local factors may influence acoustic sampling. We determined that variation in acoustic estimates of migration activity was associated with cloud cover, crosswind, date, migrant height, migrant speed, moon illumination, tailwind, and time of night. For radar, we found crosswind, date, migrant height, migrant speed, tailwind, and time of night to be associated with variations in estimates of migration activity. Overall, our findings support those of previous studies from eastern North America and demonstrate that, despite regional differences, estimates from each method are also correlated in western North America. Our findings provide new insight into factors associated with variation in estimates of migration activity from 2 widely used methods and an improved understanding of factors that impact migration behavior.

Keywords: aeroecology, bioacoustics, bird migration, migration behavior, NEXRAD, nocturnal flight call, weather radar

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LAY SUMMARY

- We generated and compared estimates of migration activity quantified by weather radar and acoustic monitoring at 50 and 55 locations at White Sands Missile Range in western North America during the fall migration seasons of 2021 and 2022.
- We found a strong positive relationship between estimates of migration activity from each method on a seasonal basis, but the relationships varied significantly from night to night.
- Variability in estimates of migration activity was associated with a variety of factors including cloud cover, crosswind, date, migrant height, migrant speed, moon illumination, and time within the night.
- Our research corroborates studies conducted in eastern North America which have found positive relationships between estimates of migration activity derived from weather radar and nocturnal flight call monitoring across nights, but not within nights.
- Additional research into the factors contributing to variability in individual flight calling behavior is needed to develop a more complete understanding of avian migration behavior.

Evaluación de métodos para estimar la actividad migratoria nocturna de aves: Una comparación entre radar y monitoreo de llamadas en vuelo nocturnas en el oeste de América del Norte

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RESUMEN

La migración de aves implica los movimientos de miles de millones de individuos, pero es difícil de estudiar porque ocurre principalmente de noche. Buscamos mejorar el entendimiento de los métodos disponibles para estudiar la migración, particularmente en regiones poco estudiadas del oeste de América del Norte. Evaluamos dos métodos: el radar meteorológico y el monitoreo de llamadas en vuelo nocturnas. Analizamos la variabilidad en las estimaciones de actividad migratoria de cada método, cómo se relacionan las estimaciones, e identificamos factores asociados con la variación en esta relación. Colectamos datos de radar y de llamadas en vuelo del sur de Nuevo México en el oeste de América del Norte durante la migración de otoño de 2021 y 2022. Estudios similares se han realizado en el este de América del Norte, pero se desconoce si la variabilidad regional altera la relación entre las estimaciones de cada método. Encontramos que las estimaciones estaban positivamente relacionadas a lo largo de la temporada, pero las relaciones variaban entre las noches. También descubrimos que la fuerza de la asociación entre los métodos variaba según los sitios, indicando que los factores locales pueden influir en el muestreo acústico. Determinamos que la variación en las estimaciones acústicas de actividad migratoria estaba asociada con la cobertura de nubes, el viento cruzado, la fecha, la altura del migrante, la velocidad del migrante, la iluminación de la luna, el viento a favor y la hora de la noche. Para el radar, encontramos que el viento cruzado, la fecha, la altura del migrante, la velocidad del migrante, el viento a favor y la hora de la noche estaban asociados con la variación en las estimaciones de actividad migratoria. En general, nuestros hallazgos respaldan los de estudios previos del este de América del Norte y demuestran que, a pesar de las diferencias regionales, las estimaciones de cada método también están correlacionadas en el oeste de América del Norte. Nuestros hallazgos brindan una nueva perspectiva sobre los factores asociados con la variación en las estimaciones de actividad migratoria de dos métodos ampliamente utilizados y una mejor comprensión de los factores que afectan el comportamiento migratorio.

Palabras clave: aeroecología, bioacústica, comportamiento migratorio, llamada en vuelo nocturna, migración de aves, NEXRAD, radar meteorológico

INTRODUCTION

Avian migration is a complex phenomenon involving seasonal movements of billions of birds across hemispheres (Dokter et al. 2018, Horton et al. 2019b). Despite the magnitude of these movements, the spectacle of avian migration is challenging to observe and document because most species primarily migrate at night. In North America, ~80% of migratory bird species migrate nocturnally (Horton et al. 2019a), representing billions of individual birds and a substantial proportion of avian biomass on the continent (Dokter et al. 2018). While standard diurnal survey methodologies, such as point counts, are not feasible at night, researchers have used alternative tools, such as acoustic recorders and weather radars, to monitor avian movements during nocturnal migration (Van Doren et al. 2023).

Many bird species vocalize during nocturnal migration with species-specific calls (Farnsworth 2005, Van Doren et al. 2024). These nocturnal vocalizations are typically single notes ranging from 50 to 300 ms in duration, occurring in the 1 to 9 kHz frequency range (Farnsworth 2005). Nocturnal flight call monitoring is particularly advantageous in migration research because calls can be identified to species, a level of taxonomic resolution that is not possible with methods such as weather radar. Since the late 1800s, ornithologists have detected and counted the flight calls given by nocturnal migrants to estimate migration activity (Libby 1899, Farnsworth 2005). Early on, nocturnal flight call monitoring was done primarily by ear or with recorders at single sites and only applied to small spatial and temporal scales. Recent advancements, however, in both microphone and autonomous recording technology (Gibb et al. 2019) and computing (Lostanlen et al. 2019, Van Doren et al. 2024), have allowed the recording and counting of nocturnal flight calls to increase in scale (Van Doren et al. 2023, 2024). Presently, nocturnal flight call monitoring through acoustic recording is being used to study migration across entire migratory seasons (Sanders and Mennill 2014, Gesicki et al. 2016, Gyekis et al. 2019, Van Doren et al. 2023), over multiple years (Smith et al. 2014), and across continents (Van Doren et al. 2024).

While nocturnal flight call monitoring has advanced rapidly as a method to estimate migration activity and study migration more generally, limitations still exist that restrict the use of the method, or bias estimates of migration activity. Specifically, acoustic recorders have a relatively small sampling range when detecting nocturnal flight calls, with some authors estimating the maximum range to be from 200 to 800 m above ground level depending on the species calling (Evans and Mellinger 1999, Evans and Rosenberg 2000, Farnsworth et al. 2004), which is below the altitude at which many individuals migrate (Horton et al. 2016b). Atmospheric conditions and local habitat may also limit the sampling range via attenuation or obstruction of sound in recordings (Horton et al. 2015b). Additionally, nocturnal flight calling behavior varies among species and with a variety of organismal (age, sex, physical condition) and environmental factors (weather conditions, artificial lighting), all of which may influence flight call counts (Morris et al. 2016, Watson et al. 2016, Tegeler et al. 2018, Ress et al. 2024). The nature and extent of these biases are poorly understood and generally unaccounted for (Farnsworth 2005), although even raw call counts contain meaningful information about migration timing and intensity (Van Doren et al. 2023).

In addition to nocturnal flight call monitoring, ornithologists have relied on weather radar as a tool to study nocturnal avian migration for more than 75 years (Brooks 1945, Lack and Varley 1945). In the continental United States, the National Weather Service, Federal Aviation Administration, and U.S. Air Force operate an array of 143 S-band doppler weather radars, collectively referred to as the Next Generation Weather Radar (NEXRAD) system (Crum et al. 1993). While the primary purpose of the NEXRAD system is weather surveillance, data collected by the system also includes information relating to volant organisms occupying aerial habitats (Diehl 2013). In recent decades, advancements in radar technology (Stepanian et al. 2016) and improved data processing methods (Nebuloni et al. 2008, Dokter et al. 2019, Lin et al. 2019) have facilitated rapid growth in the field of aeroecology (Chilson et al. 2012). Data from the NEXRAD system has been used widely to monitor nocturnally migrating birds at both small (Van Doren et al. 2017) and large (Van Doren and Horton 2018, Dokter et al. 2018, Horton et al. 2023) spatial and temporal scales. As with nocturnal flight call monitoring, the use of weather radar data to study nocturnal avian migration has limitations that may restrict its use. Unlike nocturnal flight call monitoring, which samples calls given by individuals that are often identifiable to species, weather radar measures total migrant biomass, which does not allow for identification at any taxonomic resolution below Class (e.g., birds vs. bats vs. insects). Additionally, the lowest sampling angle for radars in the NEXRAD system is ~0.5°, meaning radars often

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miss birds migrating at lower altitudes relative to ground level (e.g., < 50 to 100 m above ground level). Last, weather radar data, in terms of its use to study bird migration, is susceptible to contamination from precipitation and non-avian organisms (such as bats and insects). When present, these contaminants can mask the signal of migrating birds; however, some effective methods do exist to remove these contaminants from radar data (Lin et al. 2019).

Previous studies in eastern North America have compared estimates of migration activity as generated by acoustic and radar approaches to illuminate how the various limitations and biases of each data type may impact these estimates (Larkin et al. 2002, Farnsworth et al. 2004, Gagnon et al. 2010, Horton et al. 2015a, Van Doren et al. 2023). These studies found that flight height negatively affected call counts, and that call counts increased throughout the night (Farnsworth et al. 2004, Horton et al. 2015a). While Farnsworth (2005) notes that call rates increase with more cloud cover, Horton et al. (2015a) found no evidence of this relationship, but a decline in call rates with increasing flight speeds. Van Doren et al. (2023) found a close correspondence between radar and call count estimates of migration activity once weather and seasonal timing were accounted for. Each of these studies has revealed generally positive relationships between estimates of migration activity as derived from radar and acoustic monitoring; however, the strengths of these relationships have been highly variable among studies. Furthermore, each of these studies noted high temporal variability in correlation between the 2 methods, both within and between nights, that remains to be fully explained. Notably, all of these studies were conducted in the eastern half of North America (La Sorte et al. 2014). The extent to which results from the particular community of avian migrants and a suite of atmospheric conditions found in the eastern half of North America are applicable to other regional migratory flyways with differing migrant communities and atmospheric conditions remains unclear.

Here we compare estimates of avian migration activity obtained using acoustic recordings and weather radar at a location in the western United States to elucidate the potential biasing factors and limitations of each method for studying nocturnal avian migration. We aimed to determine (1) what factors are associated with variability in estimates of migration activity for each method in this region, (2) how estimates of migration activity from the 2 methods relate with each other, and (3) what factors are associated with the variability in the relationship between these 2 methods. We sampled 2 fall migration seasons in southern New Mexico, USA using a network of acoustic recorders and a nearby NEXRAD site to test the strength of correspondence between estimates of migration activity via call rate (as sampled by acoustic recording) and reflectivity rate (as sampled by weather radar). Additionally, we modeled the variation in acoustic and weather radar-based estimates of migration activity, and the variability in the relationship between the 2 using a suite of climactic variables and variables intrinsic to migration activity (such as flight speed and height). Based on previous results, we predicted call counts would increase with increasing birds aloft, increased cloud cover, decreased flight height, and increase over the course of a night. For migration activity rates derived from weather radar, we predicted activity rates would be highest in the hours immediately following sunset, would increase on nights with favorable migration conditions (increased tailwind, reduced cloud cover), and would increase on nights with higher average migrant speeds. We predicted that estimates of migration activity from both methods would be highly variable on a night-to-night basis. We also predicted a generally positive relationship between estimates derived from the 2 methods, with the strength of this relationship varying across nights. Finally, we expected stronger associations on nights with lower average migrant flight height, as a higher proportion of migrants would be detected by ground-based acoustic sensors.

METHODS

Study Area

We quantified avian migration activity using acoustic and weather radar methods within the Tularosa Basin of southern New Mexico, USA (Figure 1). The Tularosa Basin, located within the Chihuahuan Desert in the southwestern United States (32.4020°N, 106.3329°W), covers ~16,800 km^2 and is bordered to the east by the Sacramento Mountains and to the west by the San Andres and Oscura Mountains. Avian migration through the Tularosa Basin is consistent with much of the western United States and distinct from that of the eastern United States in species composition and density (La Sorte et al. 2014). Migrant densities are comparatively low in the West relative to the East, and many species that pass through the eastern United States in great numbers (such as *Catharus ustulatus* [Swainson's Thrush] and *Setophaga ruticilla* [American Redstart]) comprise only a small proportion of the migrant community in our region.

Acoustic Data Collection and Processing

We recorded nocturnal flight calls using passive acoustic recorders (Wildlife Acoustics Song Meter Micro, Maynard, Massachusetts, USA; 44,100 Hz sampling rate, 12 dB gain, 16-bit.wav file format) across ~550 km^2 in White Sands Missile Range, New Mexico, during fall migration. Recordings were made at 50 locations in 2021 and 55 in 2022 (Figure 1). Recorders operated nightly from 1.5 hr after sunset to 1.5 hr before sunrise from September 1st to November 1st, resulting in 55,593 hours of recordings (25,726 in 2021 and 29,867 in 2022). The nightly temporal sampling window was chosen to reduce the recording of vocalizations from birds on the ground and not in active migration, the inclusion of which could bias call counts. We chose to sample during fall migration as the density of migrants passing through our study region is higher in the fall than in the spring. Also, we chose the date range of September 1st to November 1st to include peak migration in the region (typically mid-September) as well as periods when activity rates are generally low (mid to late October). Recorders were mounted ~1.5 m high on metal posts to minimize vegetation interference.

We used Nighthawk, available on GitHub (https://github.com/bmvandoren/Nighthawk/), to detect nocturnal flight calls within recordings (Van Doren et al. 2024). The core Nighthawk model (version 0.2), which was trained in part using 45 hr of audio collected from our study area, was able to successfully detect 69% of manually annotated flight calls within the 45 hr of audio from our study area with a false positive rate of 3.5%. To maximize detection rates for our study area, we fine-tuned the core Nighthawk model

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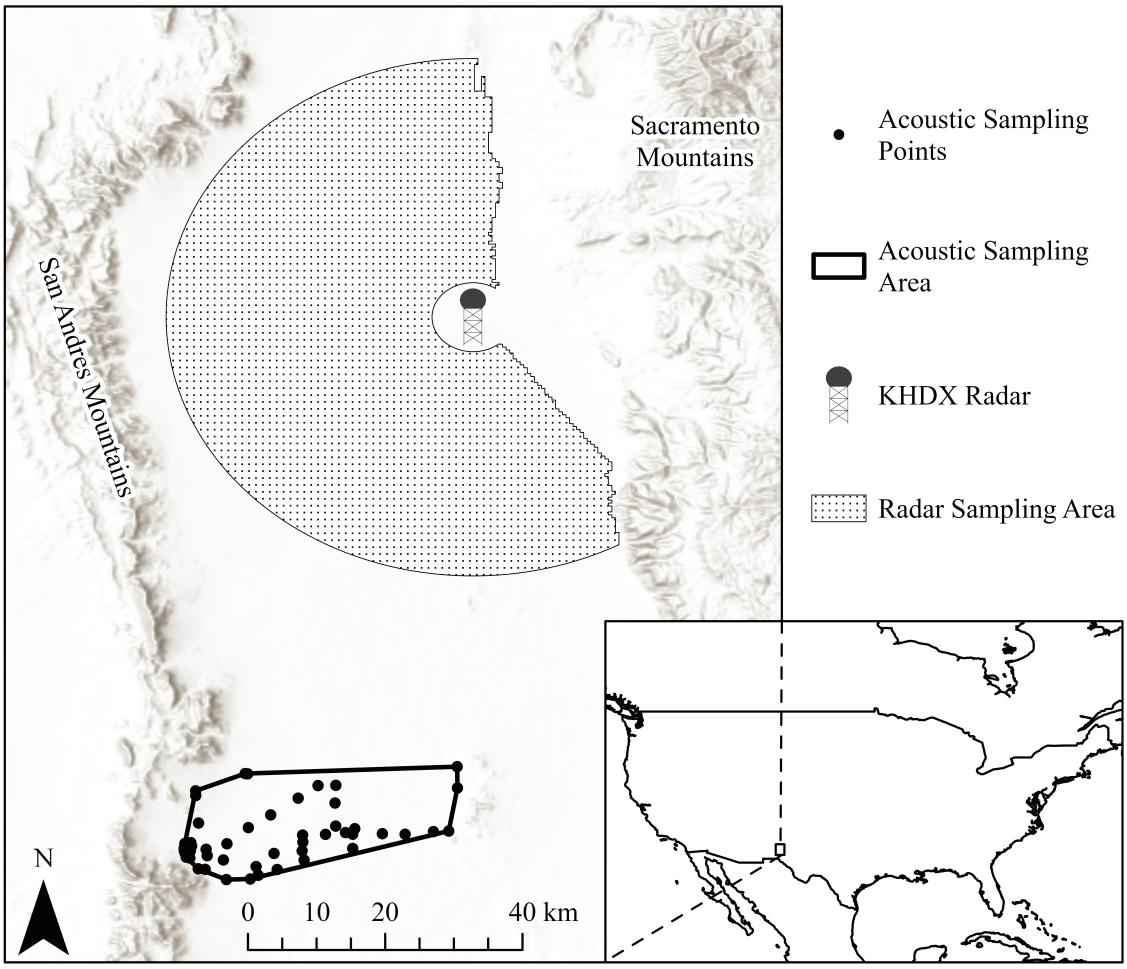

FIGURE 1. Map of study area depicting locations of radar and acoustic recorder sampling areas in southern New Mexico, USA 2021 to 2022. Radar sampling was conducted within a range of 5 to 37.5 km from the KHDX radar site for all areas in which the radar beam was not blocked by topography or structures. While sampling areas do not directly overlap, they lie directly along an expected north to south migration corridor through the Tularosa Basin.

using ~700 hr of recordings collected within our study area (~500 hr from 2021 and ~200 hr from 2022) (Van Doren et al. 2024). We then used this fine-tuned version of Nighthawk to detect flight calls within our recordings, retaining all detections with probability values >0.5 and retained only calls from Order Passeriformes (representing 98% and 97% of all calls in 2021 and 2022, respectively) as classified by Nighthawk, given that the fine-tuning data were comprised almost entirely of Passeriformes flight calls. To verify the performance of the fine-tuned model on our data, we compared Nighthawk's output with manual call counts from 100 randomly selected 15-min recording segments and found a strong correlation (r = 0.92), demonstrating that data generated by the fine-tuned version of Nighthawk are an accurate index for nocturnal flight call rates (Supplementary Material Figure 1).

We divided each night with any acoustic recordings into 10 equal temporal segments (deciles) and then calculated a mean call rate (calls hr^{-1}) for each recording site during each decile on each night. Decile 1 began when recordings began (1.5 hr after sunset) and decile 10 ended when recordings ended (1.5 hr before sunrise). We chose this temporal sampling window to limit the number of vocalizations recorded from birds on the ground and not in active migration. Decile length ranged from 0.84 hr to 1.03 hr, corresponding with changes in night length.

Weather Radar Data Collection

For comparison with acoustic data, we estimated bird-migration activity via reflectivity using unfiltered (level II) NEXRAD system data (Van Doren and Horton 2018). Migration activity from reflectivity has been demonstrated to correlate with the number of birds aloft as sensed via acoustics and thermal imaging for migrants in eastern North America (Horton et al. 2015a) and is commonly used as a metric to assess avian migration activity (Stepanian et al. 2016). Specifically, we used the bioRad package (Dokter et al. 2019) in R (R Core Team 2023) to access and download data for the KHDX radar site from the Amazon Web Services repository (https://s3.amazonaws.com/noaa-nexrad-level2/index.html). The KHDX radar is located within the Tularosa Basin near Tularosa, New Mexico, USA, 75 km from the center of our acoustic recording sites, and 65 and 87 km from the closest and farthest acoustic recording sites, respectively (Figure 1). At this range, areas sampled by acoustic recorders and weather radar do not directly overlap, with the furthest extent

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of the radar sampling range that we analyzed being ~37.5 km from the center of our acoustic recording sites (Figure 1). While the sampling areas used in this study do not directly overlap, they are in close proximity along the expected north-south flight path of migrants in this region. Additionally, local topographic features (such as the San Andres and Sacramento Mountains; Figure 1) may create a corridor for migratory birds through the Tularosa Basin along this same north to south path. Therefore, we expected that most migrants passing through the acoustic sampling area also passed through the radar sampling area. We downloaded all radar scans made between September 1st and November 1st, in 2021 and 2022. From the downloaded data, we retained all radar scans that were collected between 1.5 hr after sunset and 1.5 hr before sunrise, matching the timespan for which acoustic data were collected (Horton et al. 2015a). While this nightly temporal sampling window may not sample birds during peak exodus (Cooper et al. 2023), it does include most nightly migration activity (Kranstauber et al. 2023).

Following the methods of Van Doren et al. (2021), we used bioRad to create vertical profiles of reflectivity for each radar scan at 100-m intervals 100 to 3,000 m above ground level using the 5 lowest elevation scans (0.5° to 4.5°) within a range of 5 to 37.5 km from the KHDX radar site. At these ranges, the height of the center of the radar beam for the lowest angle scan (0.5°) is ~45 m and 410 m above ground level at 5 km and 37.5 km from the radar location, respectively. Additionally, for the lowest angle scan, the radar beam covers an altitudinal range of 1 to 89 m at 5 km from the radar site and 83 to 737 m at 37.5 km from the radar site. We then integrated reflectivity across these vertical profiles using the *integrate_profile()* function in *bioRad*. Prior to integration, we removed precipitation and retained biological targets from the unfiltered data by eliminating data with correlation coefficient > 0.95 or reflectivity factor > 35 (Horton et al. 2016a, 2018, 2020). Additionally, we removed data in which biological targets were moving in the direction opposite of the expected migratory direction for our area (i.e., azimuths between 90° and 270° were retained), as these data likely represent insect and not avian migrant-dominated scans (Horton et al. 2019b, 2020).

For each scan we extracted an estimate of vertically integrated reflectivity (in $cm^2\ km^{-2}$), flight height (in m above sea level; calculated as mean flight height weighted by reflectivity), horizontal ground speed of migrants (in m s^{-1}), and direction of horizontal ground speed (in degrees). We then calculated the mean reflectivity rate, flight height, and ground speed of migrants within each decile of each night that acoustic data were collected.

Covariate Data Collection

We used weather covariates of cloud cover, moon illumination, wind direction, wind speed, crosswind and tailwind, and temporal covariates of decile and night, to investigate factors driving variability in call rate, reflectivity rate, or variability in the relationship between these 2 metrics of migration activity. Cloud cover data were sourced from the Holloman Air Force Base weather station (52 km from the center of our recording locations, 17 km from the KHDX radar), classified on a scale of 1 (no cloud cover) to 5 (complete cloud cover) and averaged within each decile for each night (Iowa Environmental Mesonet 2023). Moon illumination data, corresponding to the proportion of the moon illuminated above the horizon, were collected for the location corresponding with the center of our recording locations for all deciles using the getMoonIllumination() and getMoonTimes() functions from suncalc in R (Thieurmel and Elmarhraoui 2022). Wind speed and direction were provided by White Sands Missile Range Meteorology Branch and were collected at a height of 10 m at a location 14 km from the center of our recording locations and were averaged within each decile for each night. From the wind data, we calculated tailwind assistance (m s^{-1}; hereafter "tailwind") as the component of the wind velocity that is along the same axis as the expected flight direction, where positive values indicate a tailwind, and negative values indicate a headwind. Additionally, absolute crosswind (m s^{-1}; hereafter, "crosswind") was calculated as the component of the wind velocity that is perpendicular to the expected flight direction. Tailwind and crosswind were calculated using the following formulas following Becciu et al. (2018):

$$\text{Tailwind} = \nu_w \times \cos\left((\theta_w + 180^\circ)\theta_b\right)$$

$$\text{Crosswind} = \text{abs}(\nu_w \times \cos\left((\theta_w + 90^\circ)\theta_b\right)),$$

where ν_w is the wind velocity (m/s), θ_w is the wind direction in degrees, and θ_b is the expected flight direction of migrants (i.e., 180° for our study).

For temporal covariates, decile corresponds to the tenth of the night within which data were collected. The night covariate corresponds to the ordinal date at sunset on the night for which the data were collected (e.g., night 244 corresponded to data collected between sunset on September 1st to sunrise on September 2nd). The year, either 2021 or 2022, corresponds to the year in which data were collected. Descriptions of all variables can be found in Supplementary Material Table 1.

Statistical Analyses

Relationship between estimates from radar and acoustics

We assessed the relationship between migration activity estimates from call counts and weather radar data for 3 comparisons for each year: (1) across nights, (2) within nights, and (3) within deciles across nights. Using linear mixed models in a hierarchical Bayesian framework, we analyzed reflectivity rate (response) against call rate (predictor), with site as a random effect (Schielzeth et al. 2020). For the across-nights comparison, we fitted 2 models (Table 1, Models I and II). Model I used mean rates within each decile per site per night, while Model II mirrored Horton et al. (2015a) by using mean rates across all deciles per night per site. Therefore, Model I addressed decile-to-decile variability, while Model II assessed overall nightly correlations. Models III and IV were fitted for the within-night comparison using mean rates within each decile for each site for each night to examine relationships within nights across sites (Model III) and across nights for each site individually (Model IV). For the within-decile comparison, we fitted Model V using mean rates for each decile for each site for each night and assessed the relationship within each decile across nights and across deciles overall. Last, to understand the effect of distance between our sampling areas (~75 km; Figure 1) on the relationship between estimates from each method, we lagged radar estimates by 2 deciles (~2 hr), which is the time it would take for a typical

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TABLE 1. Descriptions of the hierarchical Bayesian linear mixed models used to assess the relationship between estimates of migration activity from weather radar and from nocturnal flight call monitoring. CPH = calls per hour. VIR = vertically integrated reflectivity.

Model [a]	Level of comparison	Parameter estimate specificity	Data structure
Model I	Across-nights	Site specific, sites overall	CPH and VIR values for each decile for each night for each site (10 total values per site per night).
Model II	Across-nights	Site specific, sites overall	Mean CPH and VIR values across deciles for each night for each site (one value per site per night).
Model III	Within-night	Night specific across sites for each night, nights overall	CPH and VIR values for each decile for each night for each site (10 total values per site per night).
Model IV	Within-nights	Site specific for each night, sites overall	CPH and VIR values for each decile for each night for each site (10 total values per site per night).
Model V	Within-deciles	Decile specific across sites, deciles overall	CPH and VIR values for each decile for each night for each site (1 value for each decile per site per night).

[a]Each model was run twice, once with concurrent data and once with time-lagged data.

migrant to cover the distance between sampling areas (Larkin 1991), and repeated the analyses above.

Prior to modeling, we square-root transformed radar estimates for normality and standardized predictor variables to have a mean of 0 and a variance of 1 using R's *scale()* function. We removed all sites for which call rate data was missing from >20% of deciles given recorder malfunctions (4 of 50 sites in 2021 and 3 of 55 sites in 2022). Last, as the models could not handle missing data, we imputed any remaining missing values with median values from the same decile on the same night calculated across all sampled sites. Following these procedures, we were left with 23,000 unique site-night-decile combinations for 2021 and 28,600 for 2022.

Bayesian models were generated in R with *rjags* (Plummer 2023). We used uninformative normal and uniform prior distributions for model parameters and tuned samplers using the first 500 samples, discarded 5,000 burn-in iterations, ran 35,000 MCMC iterations, and saved every fifth iteration (Bolker 2008). We monitored convergence using R-hat values and trace plots, increasing iterations and burn-in as needed. We reported mean estimates and 85% credible intervals for each parameter of interest (Arnold 2010, Sutherland et al. 2023). Full model specifications can be found in the code files provided.

Factors associated with migration activity and variability in the relationship between methods

We analyzed factors influencing variability in migration activity estimates using a priori linear mixed-effects models. We used log-transformed call rate (calls/hr; 36 models) for each decile for each site, or square-root transformed mean reflectivity rate (cm^2 km^{-2}; 35 models) for each decile as the response variable and cloud cover, moon illumination, tailwind, crosswind, migrant flight height, migrant flight direction and average migrant speed, decile, night, and year as predictors and site as a random effect for models with call rate as the response, and night as a random effect for both models (Supplementary Material Tables 2 and 3). Model sets included a univariate model for each predictor, a global model with all predictors, a null model, and additional models based on anticipated and previously discovered relationships between predictor variables and migration activity estimates as summarized in the Introduction section. For models (other than univariate models) with call rate as the response variable, we also included mean reflectivity rate as a predictor. Last, to analyze potential contributors to the variability in the relationship between nocturnal flight calls and weather radar-derived estimates of migration activity between nights, we used the set of a priori models as above, with coefficient values from the within-nights comparison (Model III) for each site for each night as the response variable (Supplementary Material Table 4).

All models were fitted and 85% confidence intervals were calculated using the lmer() function in the lme4 package in R on scaled and centered variables (Arnold 2010, Bates et al. 2015), and compared by Akaike's information criterion for small sample sizes (AIC_c; Burnham and Anderson 2004). All analyses were performed and figures were generated using R through the New Mexico State University High Performance Computing group (Trecakov and Von Wolff 2021).

RESULTS

We recorded 25,726 hr of acoustic data in 2021 and 29,867 hr in 2022. Within these recordings, our fine-tuned Nighthawk model detected 673,520 flight calls in 2021 and 1,289,177 calls in 2022 (Figure 2). Average call rate was lower during the 2021 season (26.19 calls hr^{-1}) than in 2022 (43.16 calls hr^{-1}). We processed and retained a total of 5,548 radar scans from 2021 and 6,227 scans from 2022. From the scans, the nightly average reflectivity rate was 257.37 cm^2 km^{-2} in 2021 and 254.66 cm^2 km^{-2} in 2022. Across both data collection periods, migration activity was at its highest rate during the first month of data collection (September), with activity rates declining throughout the remainder of the period. Over the course of an average night, estimates derived via acoustic monitoring increased while estimates from weather radar decreased (Figure 3). These patterns were generally consistent between 2021 and 2022, with a stronger trend in the radar data in 2021 than in 2022.

Relationship Between Radar and Acoustics

For the across-nights comparison (2,300 unique site-night combinations for 2021 and 2,860 for 2022), in which we averaged estimates from each method within each night for each site, we found a positive relationship between estimates of migration activity as detected via weather radar and acoustic monitoring for 2021 (coefficient

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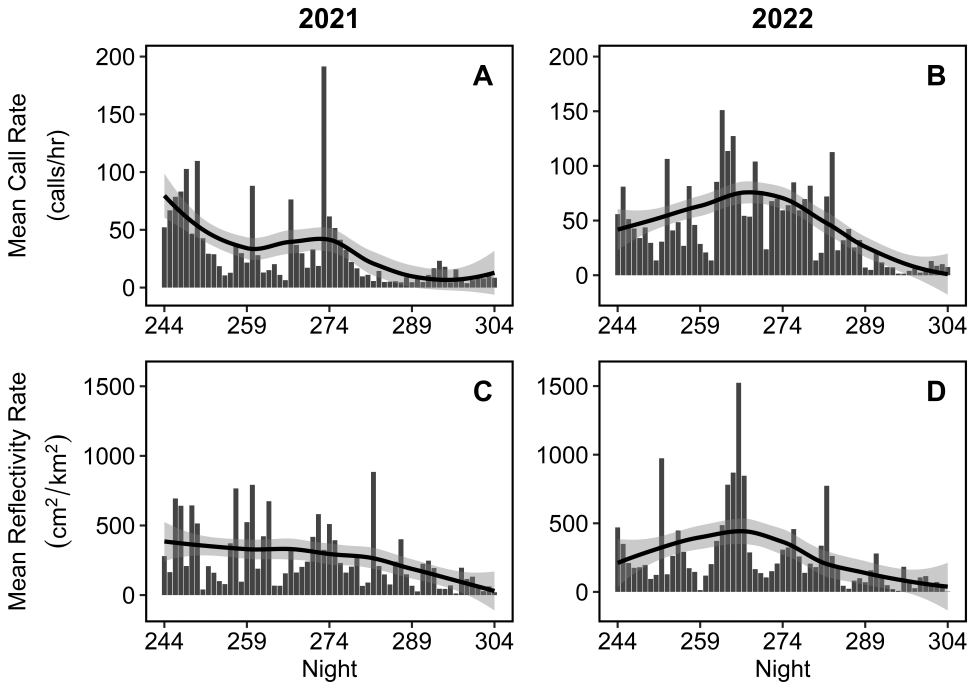

FIGURE 2. Mean nightly migration activity estimates in south central New Mexico for fall migration 2021 and 2022 as derived from (**A, B**) nocturnal flight call monitoring, and (**C, D**) weather radar. Curves presented are generated using the loess method and include 85% confidence intervals. Night 244 corresponds with the night of September 1.

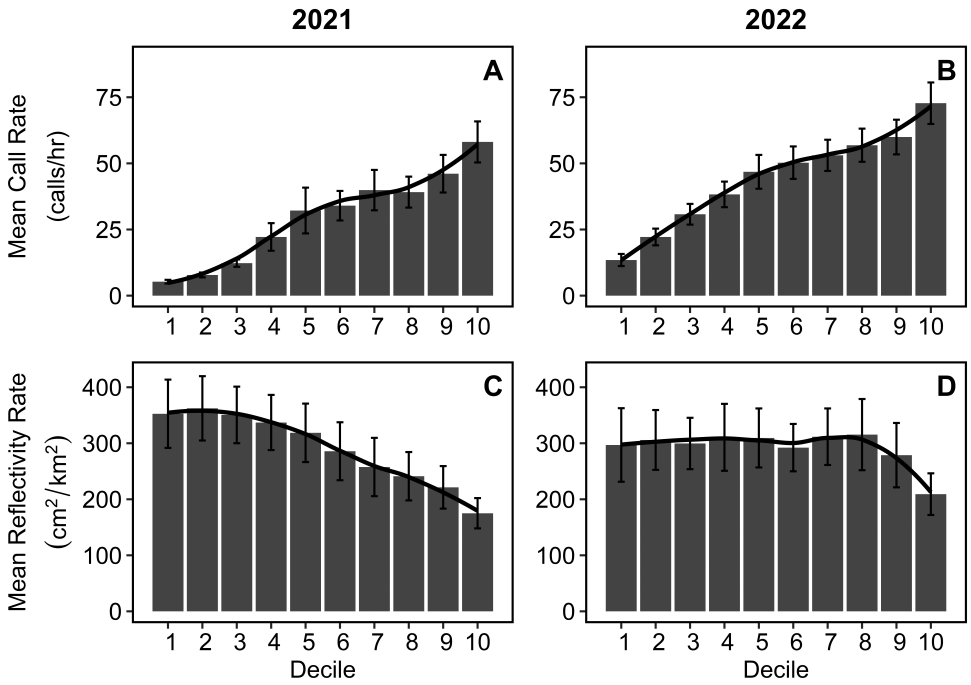

FIGURE 3. Mean migration activity estimates by decile for 2021 and 2022 as derived from (**A, B**) nocturnal flight call monitoring, and (**C, D**) weather radar in southern New Mexico. Bars and whiskers correspond with mean ± SE. Curves, generated using the loess method, are presented to emphasize trends. Decile 1 began 1.5 hr after sunset and Decile 10 ended 1.5 hr before sunrise.

estimate = 0.34, 85% CI = 0.28 to 0.40), and 2022 (coefficient estimate = 0.42, 85% CI = 0.39 to-0.45) (Table 2, Figure 4A,B; Supplementary Material Table 5). For the across-nights comparisons where estimates were analyzed at the decile level, rather than averaged within each night at each site, weaker relationships were detected across nights for 2021 (coefficient estimate = 0.19, 85% CI = 0.16 to 0.22), and for 2022 (coefficient estimate = 0.28, 85% CI = 0.26 to 0.31) (Table 2, Figure 5A,B; Supplementary Material Table 5).

For the within-night comparison (2,300 unique site-night combinations for 2021 and 2,860 for 2022), we found no

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TABLE 2. Summarized coefficient estimates for the anlaysis of the relationship between estimates of migration activity as obtained via acoustics and weather radar in southern New Mexico during September and October 2021 and 2022. Significant estimates, those with 85% credible intervals (which can be found in Supplementary Material Table 5) not overlapping zero, are in bold.

	2021		2022	
	Not lagged	Lagged	Not lagged	Lagged
Period	Coefficient estimate	Coefficient estimate	Coefficient estimate	Coefficient estimate
Across nights [a]	**0.34**	**0.51**	**0.42**	**0.42**
Across nights [b]	**0.19**	**0.28**	**0.28**	**0.40**
Within nights	–0.04	0.03	–0.02	**0.09**
Decile 1	**0.33**	–	**0.26**	–
Decile 2	**0.32**	–	**0.37**	–
Decile 3	**0.31**	**0.53**	**0.34**	**0.35**
Decile 4	**0.31**	**0.45**	**0.25**	**0.23**
Decile 5	**0.30**	**0.37**	**0.16**	**0.21**
Decile 6	**0.28**	**0.30**	**0.21**	**0.24**
Decile 7	**0.23**	**0.27**	**0.31**	**0.29**
Decile 8	**0.26**	**0.29**	**0.35**	**0.31**
Decile 9	**0.25**	**0.25**	**0.31**	**0.38**
Decile 10	**0.18**	**0.26**	**0.26**	**0.43**

[a]Calculated using migration activity estimates averaged within each night at each site.
[b]Calculated using migration activity estimates averaged within each decile at each night at each site.

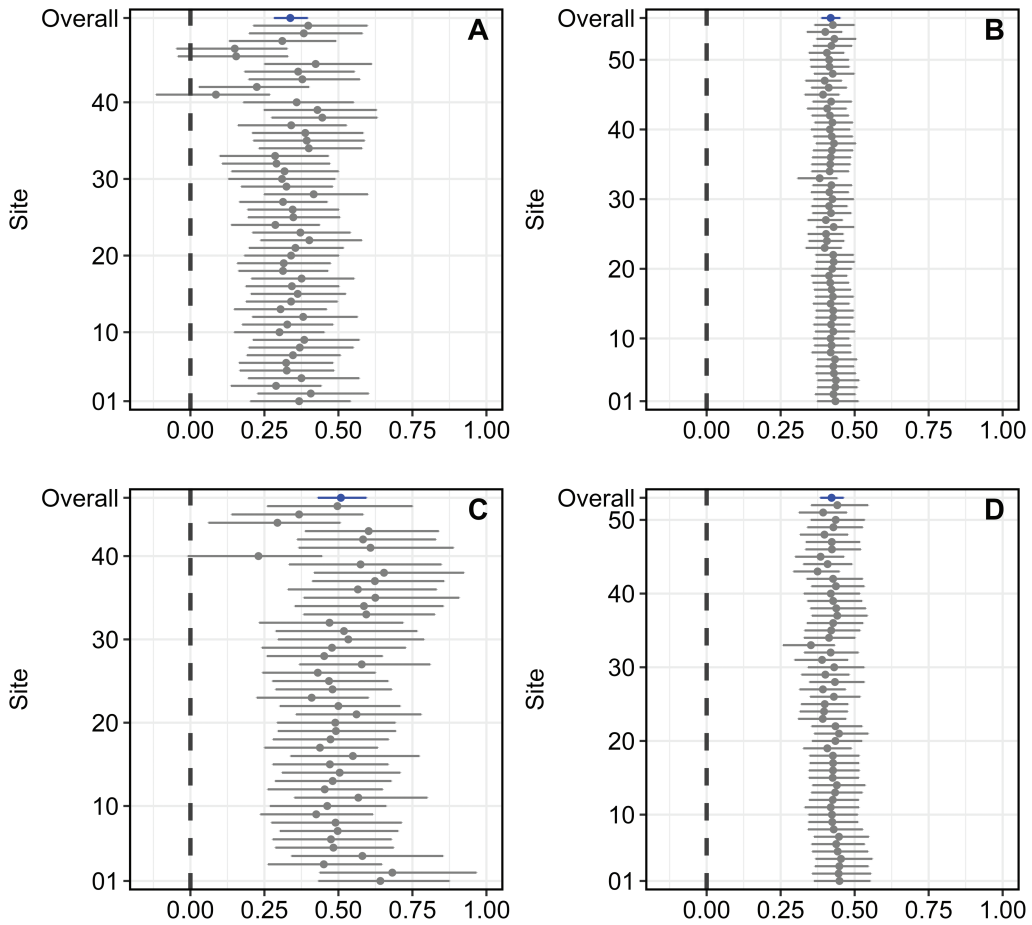

FIGURE 4. Mean estimates and 85% credible intervals of the relationship across nights between mean nightly estimates (calculated by averaging across deciles within each site for each night) of migration activity as measured by weather radar and nocturnal flight call monitoring at each site (in gray) and for sites overall (in blue) in southern New Mexico during September and October of (**A**) 2021, (**B**) 2022, (**C**) 2021 with lagged data, and (**D**) 2022 with lagged data.

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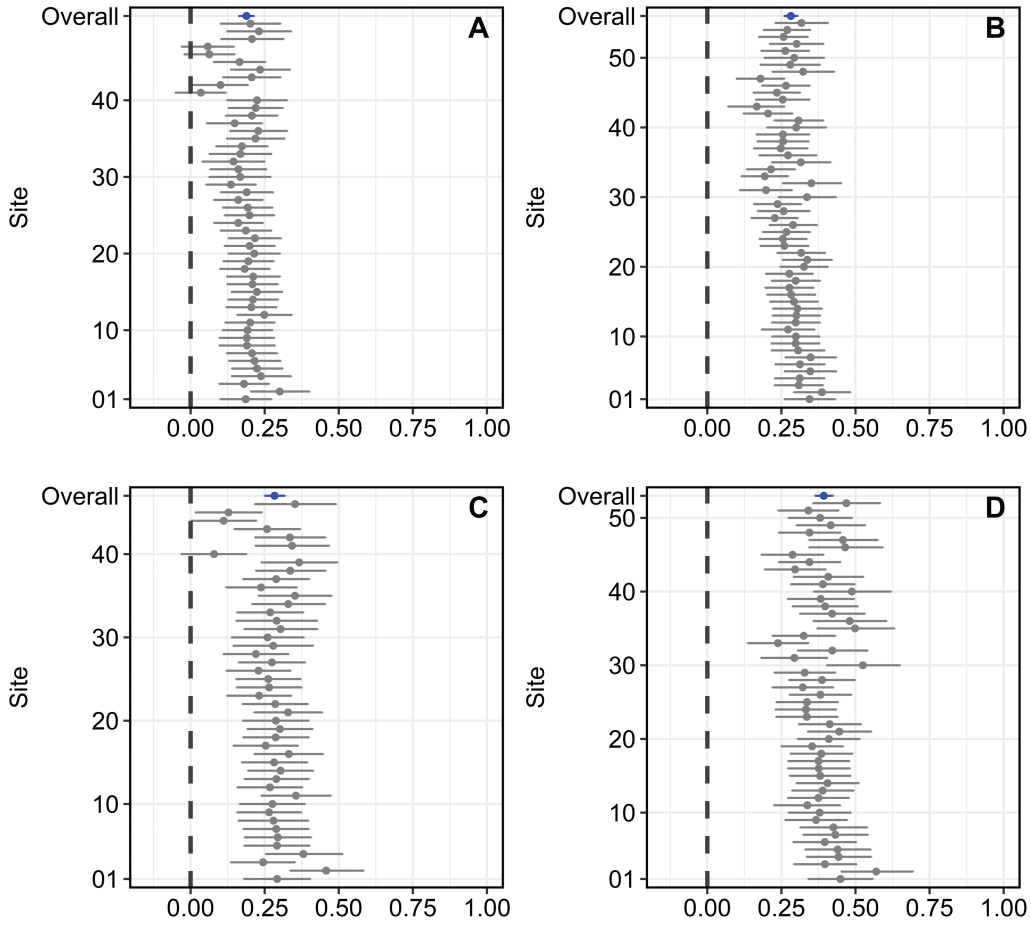

FIGURE 5. Mean estimates and 85% credible intervals of the relationships across nights between migration activity estimates as measured on a decile-by-decile basis by weather radar and nocturnal flight call monitoring at each site (in gray) and for sites overall (in blue) in southern New Mexico during September and October of (**A**) 2021, (**B**) 2022, (**C**) 2021 with lagged data, and (**D**) 2022 with lagged data.

consistent relationship between activity rate estimates from acoustics and radar for 2021 (coefficient estimate = –0.04, 85% CI = –0.09 to 0.01) and 2022 (coefficient estimate = –0.02, 85% CI = –0.12 to 0.12) (Table 2, Figure 6A,B; Supplementary Material Table 5). While significant relationships did occur within many nights, these relationships varied widely from night to night (Figure 6A,B). Relationships for the within-deciles across-nights comparisons (23,000 unique site-night-decile combinations for 2021 and 28,600 for 2022) were all positive for each year (Table 2, Figure 7; Supplementary Material Table 5).

When analyzed using lagged reflectivity data, the relationship between migration activity estimates when assessed at the decile level and the nightly level for 2021 was stronger for the across-nights comparisons (Table 2, Figure 4C,D; Supplementary Material Table 5). Using lagged reflectivity data for the across-nights analysis using nightly averaged migration estimates did not change the relationship (Table 2, Figure 4D; Supplementary Material Table 5). Similarly, for the within-night comparison, lagging the reflectivity data created small positive changes in the relationship in estimates for 2021 and 2022 (Table 2, Figure 6C,D; Supplementary Material Table 5). For the within-decile comparison with lagged data, which allowed for comparisons in deciles 3 to 10, the relationship in estimates from each method became stronger within each decile, with some variability (Table 2, Figure 7; Supplementary Material Table 5).

Factors Associated With Variation in Mean Call Rate

Of the 36 a priori models constructed to explain variation in mean call rate, the global model containing all predictors was the only competitive model (Δ AIC$_c$ < 2.0), and received all of the model weight (w_i) (Table 3; Supplementary Material Table 2). Modeling results indicated that call counts decreased with increasing cloud cover, crosswind, average migrant height, and later time of season, and increased throughout each night, and with increasing moon illumination, tailwind, and average migrant speed (Figure 8A). Modeling results also confirmed that call rate was higher in 2022 than it was in 2021 (Figure 8A).

Factors Associated With Variation in Mean Reflectivity Rate

Of the 35 a priori models constructed to explain variation in mean reflectivity rate, the global model containing all predictors was the only competitive model, and received all of the model weight (w_i) (Table 3; Supplementary Material Table 3). Model output indicated that estimates of migration activity via reflectivity decreased with the increasing tailwind, average migrant height, decile, and later time of season, while increasing with average migrant speed and crosswind (Figure 8B).

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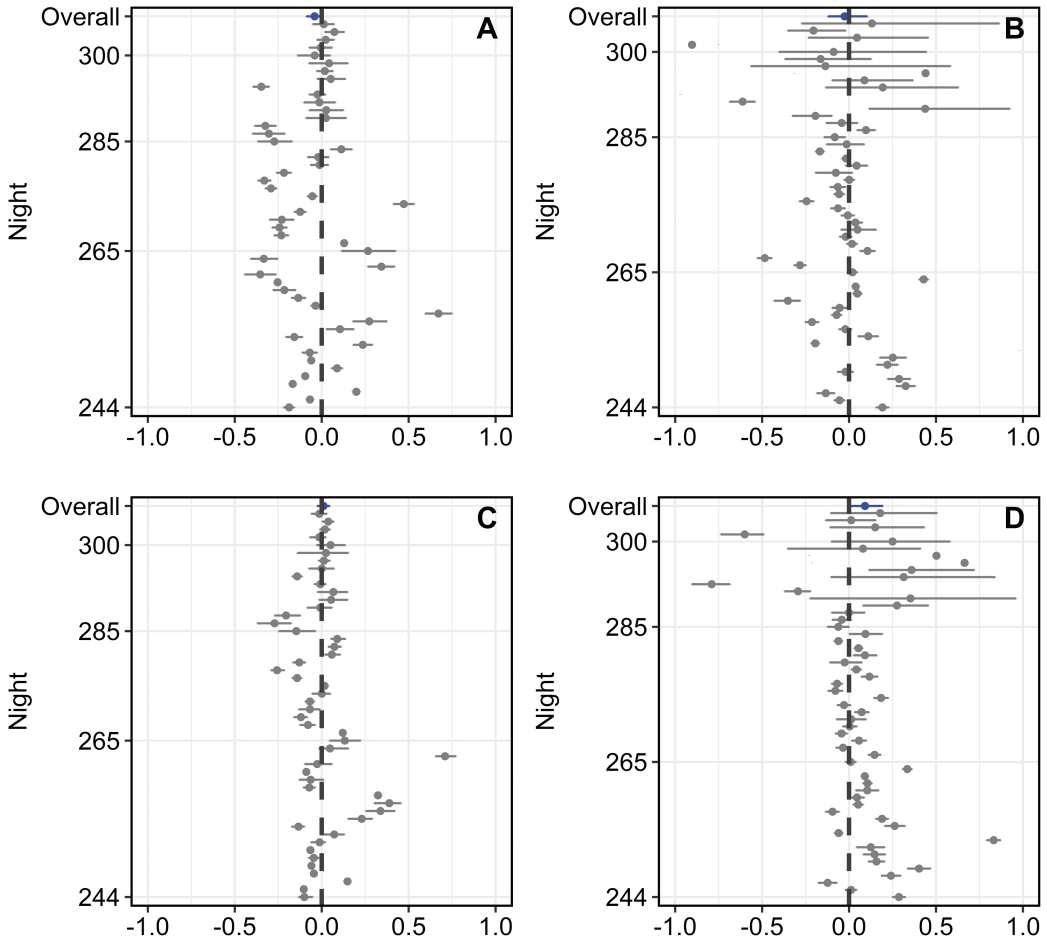

FIGURE 6. Mean estimates and 85% credible intervals of the relationships within nights across sites between migration activity estimates as measured by weather radar and nocturnal flight call monitoring and associated 85% credible intervals for each night (in gray) and for nights overall (in blue) in southern New Mexico during September and October of (**A**) 2021, (**B**) 2022, (**C**) 2021 with lagged data, and (**D**) 2022 with lagged data.

Factors Associated With Variation in the Relationship Between Radar and Acoustics

Of the 37 a priori models that we constructed to explain variation in the relationship between estimates of migration activity as detected via weather radar and acoustic recording, the global model containing all predictors was the only competitive model, and received all of the model weight (w_i) (Table 4; Supplementary Material Table 4). The model output indicated that the strength of the relationship decreased with increasing call rate, cloud cover, moon illumination, and average migrant speed, and increasing with increasing reflectivity and tailwind (Figure 8C). Additionally, the interaction between call rate and reflectivity was a significant predictor of the relationship between estimates of migration activity from radar and acoustics where the relationship became increasingly negative as call rate increased and reflectivity rate decreased (Supplementary Material Figure 2). Last, the relationship between estimates from each method was stronger in 2022 than in 2021 (Figure 8C).

DISCUSSION

Acoustic recording of nocturnal flight calls and analysis of weather radar data are widely used approaches for estimating nocturnal migration activity. For researchers to use these methods most effectively, it is important to understand what factors are associated with variability in estimates from each method, and how the relationship between estimates varies across time and space. In this study, we found that the relationship between estimates of migration activity from each method was generally positive across the migration season, indicating that both methods provide accurate estimates for migration activity at large temporal scales. Conversely, we found the relationship in estimates from each method to be highly variable on a night-to-night basis, demonstrating that within a single night, factors other than the number of birds aloft alter migration activity estimates from the 2 methods. Nevertheless, estimates from each method varied among deciles of the night, with stronger relationships occurring within deciles closest to dawn in 2022. Our findings corroborate those of previous studies conducted in eastern North America, which also found generally positive, but variable correlations between migration activity estimates as derived via acoustics and weather radar (Farnsworth et al. 2004, Gagnon et al. 2010, Horton et al. 2015a).

We sought to determine not only what causes variation in the relationship between the 2 methods, but what creates variation in estimates from each of the methods themselves. We identified a suite of variables that were considered important predictors of estimates of migration activity for each method, as well as of the strength of the relationship between

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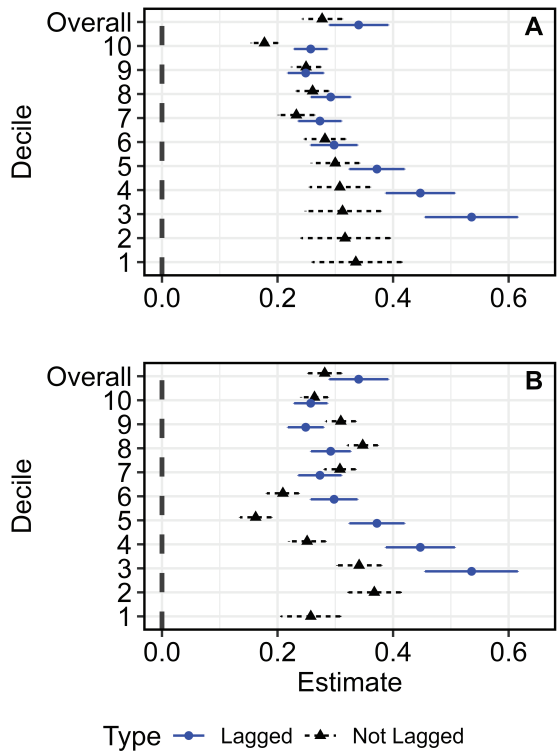

FIGURE 7. Mean estimates and 85% credible intervals of the relationships between time-lagged (in blue) and not lagged (in black) migration activity estimates as measured by weather radar and nocturnal flight call monitoring and associated 85% credible intervals for the within deciles comparison for each decile across sites and for deciles overall in southern New Mexico during September and October of (**A**) 2021 and, (**B**) 2022.

the 2 methods. Our findings provide important information regarding the variables that may create variability in estimates of activity for each method. Below, we investigate each of these variables in greater detail and discuss our findings in light of the findings of previous studies in the field.

Factors Impacting Migration Estimates via Acoustics

We found that activity estimates derived from acoustic recordings across the migration season were highly variable from night to night, which reflects the expected high variability in nightly migration passage (Farnsworth et al. 2004, Van Doren et al. 2024). Within nights, our finding that call rate increases throughout the deciles of the night, to a maximum just before dawn, is similar to what has been demonstrated in past studies in eastern North America (Farnsworth et al. 2004, Horton et al. 2015a). We found that the overall call rate was lower in 2021 than in 2022, but that this trend was not explained by variability in the number of birds aloft as reflectivity was similar between years, demonstrating that factors other than the number of migrants aloft create variability in call rate. This trend could be partly explained by the variation in the species composition of migrants passing through our study area between the 2 years (possibly driven by large-scale events, such as wildfires, droughts, or monsoons). However, this would assume that call rates vary between species of migrants, an assumption that has not been well studied. Further work is needed to determine how call rates may vary between species, and how migrant communities may vary between years within the same study area.

As predicted, we found that as migrant height increased, estimates of migration activity via acoustic recording decreased. As migrant height increases, more birds are flying above the sampling range of the acoustics recorders and are thus only detected by radar. The low-altitude sampling range of acoustic recorders may bias flight call counts towards species that typically migrate at lower altitudes, or for nights when weather conditions (e.g., low cloud ceiling, increased cloud cover) cause birds to migrate closer to ground level (Farnsworth et al. 2004, Horton et al. 2015a). An improved understanding of how each species is using the airspace during migration and what factors are associated with variation in migrant altitude may allow researchers to account for this potential bias. While multiple studies (e.g., Farnsworth 2005, Smith et al. 2014) have found that call rates increase with increasing cloud cover, others (e.g., Horton et al. 2015a) found no relationship between call rates and cloud cover. We determined that cloud cover was a significant predictor of call rates with rates decreasing with increasing cloud cover. In our study region, cloud cover is typically low and nights with high cloud cover are rare, suggesting the negative relationship we detected should be viewed with caution.

Artificial light at night is typically associated with increased call rate (Van Doren et al. 2017, Gillings and Scott 2021). In our study, we found that the call rate increased with increasing moon illumination. We are unaware of previous studies that have assessed the relationship between nocturnal flight call rates and moon illumination and believe that our finding of a positive relationship between these two variables is novel. While this finding is in line with the relationship between call rates and artificial light at night, moon illumination during our study period was not evenly distributed, with many deciles having zero moon illumination. Given that moon illumination was positively associated with call rate, but not with migration activity from radar, it is likely that the increased call rate is due to behavioral alteration and not an increase in the number of birds aloft. A potential explanation of the relationship between moon illumination and call rate may be that, as moon illumination increases, the ground surface and other obstacles become more visible to migrants, allowing for migrants to fly at lower heights above ground level (and in closer proximity to autonomous recording units (ARUs), explaining the increase in call rates). Further sampling of flight call rates under varying levels of moon illumination will aid in the understanding of how natural light at night may alter flight calling behavior.

Our modeling results indicated that crosswind is a significant predictor of calling rates, with rates decreasing with increasing crosswind. A possible explanation for the observed relationship is that as crosswind speeds increase, flight in the preferred direction may become more taxing, and a trade-off may exist between energy devoted to flight versus calling. As such, we would expect call rate to increase with increasing tailwind, which is the relationship that our analyses determined. Last, we determined that average migrant speed was a significant predictor of calling rates, with calling rates increasing as average migrant speed increases, possibly as the result of more migrants passing the recording location in a shorter time. The ability to track individual migrants in flight and analyze calling rates as a function of flight speed would allow for a more detailed examination of this trend. Furthermore, the relationship between calling rate and the weather variables studied here may be species-specific, necessitating species-specific studies of calling rates.

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TABLE 3. Mixed-effect linear regression models from the model sets testing factors associated with variation in estimates of migration activity derived from acoustic recording of nocturnal flight calls at White Sands Missile Range, and from reflectivity via weather radar from the KHDX NEXRAD in southern New Mexico during September and October 2021 and 2022. All competitive models, the first non-competitive model, and the null model are presented here for each model set; full model sets and output can be found in Supplementary Material Tables 2 and 3.

Data	Model [a]	*k*	Deviance	ΔAIC_c	w_i
Acoustic	VIR + CLD + Moon + Tailwind + Crosswind + AvgSpd + AvgHgt + CLD * Moon + Decile + Night + Year + (1 | Site) + (1 | Night)	15	125,447.44	0	1
	VIR + CLD + Crosswind + AvgSpd + AvgHgt + Decile + Night + Year + (1 | Site) + (1 | Night)	12	125,465.83	12.39	0
	Null	4	202,272.50	76803.05	0
Weather Radar	CLD + Moon + Tailwind + Crosswind + AvgSpd + AvgHgt + CLD * Moon + Decile + Night + Year + (1 | Night)	13	5137.32	0	1
	CLD + AvgSpd + AvgHgt + Decile + Night + Year + (1 | Night)	9	5268.25	122.7	0
	Null	3	7822.76	2665.01	0

[a]A plus (+) sign indicates an additive effect while an asterisk (*) denotes an interactive effect. Variable definitions are as follows; VIR = mean reflectivity rate, CLD = cloud cover, Moon = moon illumination, Tailwind = tailwind, Crosswind = crosswind, AvgSpd = average migrant speed as sensed by weather radar, AvgHgt = average migrant height as sensed by weather radar, Decile = decile, Night = night, Year = year. All continuous covariates were scaled and centered (mean = 0, SD = 1.0) prior to analysis.

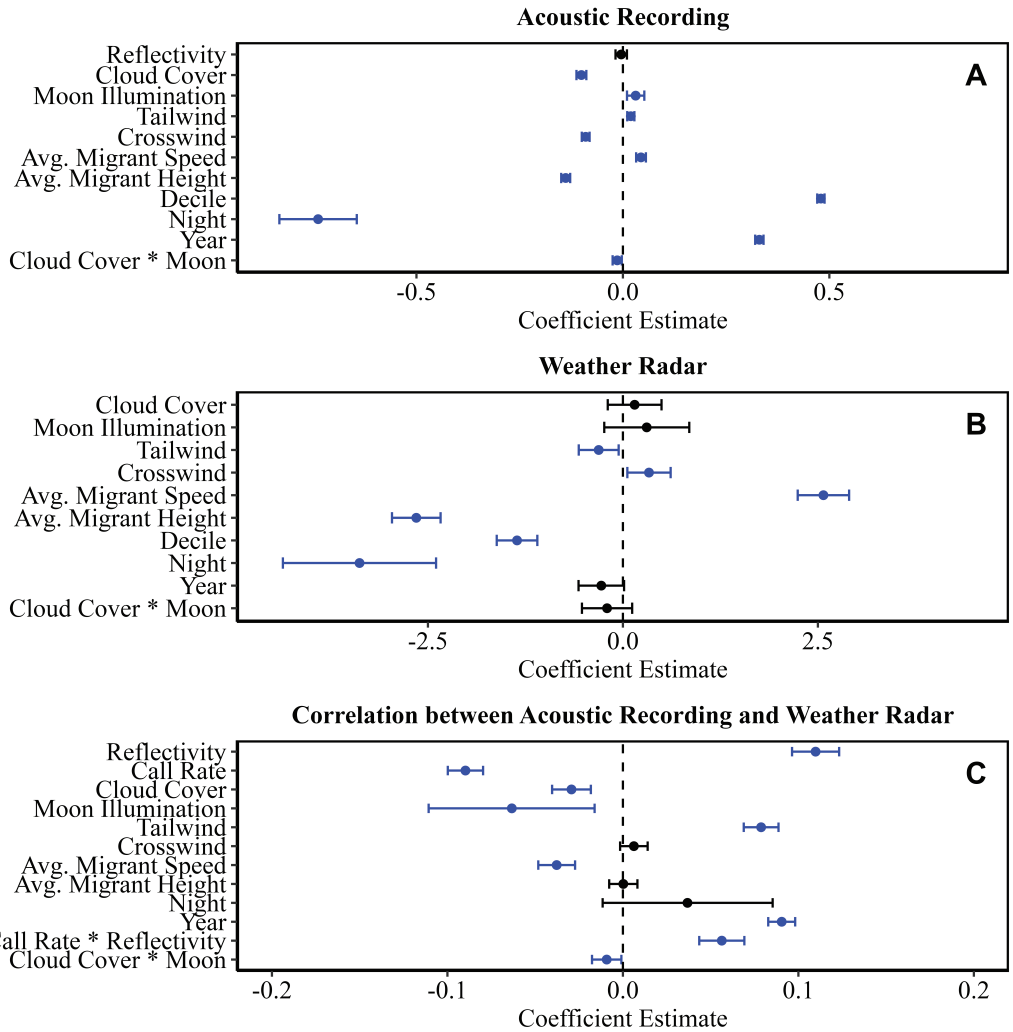

FIGURE 8. Effect size plots for the individual top model from (**A**) model set analyzing variation in migration activity rate as measured via acoustic recording; (**B**) model set analyzing variation in migration activity rate as measured via weather radar; and (**C**) model set analyzing variation in the relationship between estimates of migration activity rate as sensed by each method. Effect sizes shown are equal to the coefficient estimate (represented by dots) with 85% confidence intervals (represented by whiskers) for each predictor with significant predictors (those with confidence intervals not overlapping zero) in blue, and non-significant predictors in black.

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TABLE 4. Mixed-effect linear regression models from the model set testing factors associated with variation in the relationship between migration activity rate estimates derived from weather radar and acoustic recording in southern New Mexico during September and October 2021 and 2022. All competitive models, the first non-competitive model, and the null model are presented here; full model sets and output can be found in Supplementary Material Table 4.

Model [a]	*k*	Deviance	ΔAIC_c	*wi*
CLD + Moon + Tailwind + Crosswind + AvgSpd + AvgHgt + CLD * Moon + Night + Year + CPH + VIR + CPH * VIR + (1 | Site) + (1 | Night)	16	−2366.0983	0	1
CLD + Tailwind + AvgSpd + AvgHgt + Night + Year + CPH + VIR + (1 | Site) + (1 | Night)	12	−2318.7522	39.29	0
Null	4	451.456908	2793.44	0

[a]A plus (+) sign indicates an additive effect while an asterisk (*) denotes an interactive effect. Variable definitions are as follows: CLD = cloud cover, Moon = moon illumination, Tailwind = tailwind, Crosswind = crosswind, AvgSpd = average migrant speed as sensed by weather radar, AvgHgt = average migrant height as sensed by weather radar, Night = night, Year = year. All continuous covariates were scaled and centered (mean = 0, SD = 1.0) prior to analysis.

Three-dimensional microphone arrays (Gayk and Mennill 2020, 2023), coupled with in-flight vocalization data collected from individual migrants via on-board recording units could be one approach to disentangle the effects of wind and other variables on flight calling behavior.

Factors Impacting Migration Estimates via Weather Radar

As expected, we found that estimates of migration activity derived from weather radar data were highly variable across nights throughout the migration season (Farnsworth et al. 2004, Horton et al. 2015a, 2021). Within nights, we found that migration activity rate was highest within the first 2 to 3 hr after sunset rather than later in the night, reflecting the mass exodus of migrants from stopover habitat early in the night, followed by a gradual decline of the number of birds aloft as migrants drop out of active migration and settle into stopover habitats for the next day.

Average migrant speed was a significant positive predictor of migration activity rate. As average migrant speed (detected by weather radar) increased, migration activity rate also increased. Nights with high rates of migration are likely characterized by conditions that are conducive to migration, such as increased tailwinds/decreased headwinds (Shamoun-Baranes and van Gasteren 2011, Chapman et al. 2016, Van Doren and Horton 2018). However, we found that increased tailwind was weakly associated with decreased migration activity. Nonetheless, on nights with conditions conducive to migration, more birds will take to the skies and be able to fly at faster speeds given the positive conditions, resulting in the relationship found here (Erni et al. 2002, Shamoun-Baranes and van Gasteren 2011). Modeling results indicated that as migrant height increased, migration activity rates decreased. Weather conditions that may cause birds to migrate at greater heights (e.g., turbulent winds at ground level) are likely to be associated with fewer birds aloft given that fewer birds migrate when weather conditions are adverse, potentially explaining the documented trend.

Factors Impacting the Relationship Between Migration Estimates via Acoustics and Weather Radar

Our findings of the relationship between estimates of migration activity from each method across nights, within nights, and within deciles mirrored those of previous studies (Farnsworth et al. 2004, Horton et al. 2015a). The similarity of our findings to those of studies in eastern North America (Farnsworth et al. 2004, Gagnon et al. 2010, Horton et al. 2015a) indicates that the relationship between estimates from either method is robust to variability in the overall composition of the migrant community, topography, weather patterns, and other factors that might result from differing geography. For each year, we found a positive relationship between migration activity estimates as derived from weather radar and acoustics across nights, but the relationship was stronger when nightly estimates were used, rather than estimates on the decile level.

While the relationship between call rate and reflectivity was generally positive across nights, this was not true for all sites (Figure 4). While nocturnal avian migration is generally considered to occur evenly across space, variations in topography and geography across the landscape can create corridors with consistently high or low activity rates or may alter the behavior of migrants passing through these areas (Williams et al. 2001, Terrill et al. 2021). For instance, if our hypothesis is true that increased moon illumination allows for migrants to fly lower as obstacles are more visible, then variability in the reflectance of ground cover or the presence of obstacles at individual sites may lead to variability in recorded call rates between sites. Given the small sampling area of a single acoustic recording location, data collected from individual locations may be biased by this variability. Consequently, estimates of migration activity based on a single or small set of acoustic recording locations may not demonstrate the same strong positive relationships to weather radar-based estimates that we found in this study. Last, while the relationship between methods was positive across nights of the migration season, the relationship was highly variable night-to-night, demonstrating that each method is differentially impacted by varying conditions on a night-by-night basis.

Each method may provide an accurate estimate of migration activity at a large temporal scale, however, at smaller time scales the conditions impacting call rate and reflectivity rate may impact each method differently, biasing estimates in different directions. For instance, the decreased relationship between estimates across nights when data are considered at the decile level and not averaged within nights is likely due to the well-documented and disparate trends exhibited by call rate (increases throughout the night) and reflectivity (decreases throughout the night) that occur throughout each night. On any given night, most individual birds take off from stopover within the first few hours following sunset (Cooper et al. 2023).

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As a result, the highest density of birds aloft within a night occurs within the first few hours after sunset, as demonstrated by reflectivity data (Figure 3C). While the highest density of birds aloft occurs early in the night, most birds at this time are in active migration and flying above the sampling range of ground-based ARUs, resulting in low recorded call rates early in the night. However, as the night progresses, birds begin to drop out of migration into stopover habitat (Ellegren 1993, Bolshakov et al. 2003), placing them in closer proximity to ground-based ARUs, leading to increases in call rates throughout the night. Call rates peak near dawn as most remaining nocturnal migrants in flight land prior to sunrise (Michalik et al. 2020, Kranstauber et al. 2023). It is important to note that our study occurred inland, in an area without large geographic barriers. In coastal locations, or other locations with large geographic barriers, the timing of migration activity within a night may vary from what was found here (Kranstauber et al. 2023). Last, as evidenced by variability between sites, local-scale factors may bias call counts at specific sites, thereby impacting the correlation between methods. Our findings indicate that for studies using acoustic methods to accurately quantify migration activity, multiple recording locations should be used and distributed throughout the study area of interest. The ability of researchers to process large amounts of acoustic data using algorithms such as Nighthawk (Van Doren et al. 2024), and the declining cost of acoustic recorders make this increasingly feasible. While each method can provide a reliable estimate of migration activity at a seasonal or nightly scale, when possible, both methods should be used in a complementary fashion given the different biases that may affect each of their estimates.

Our analysis of the drivers of the relationship between estimates from weather radar and acoustics revealed multiple important variables. First, we found that as reflectivity increases, and call rate decreases, the correlation in estimates from both methods increases. This finding suggests that call rate may be overestimating migration activity, while reflectivity is possibly underestimating. As we did not attempt to exclude calls produced by the same individual from our recordings, it is not unexpected that the call rate may be overestimating migration activity as individuals may be counted multiple times. Surprisingly, even though migrant height had a significant negative relationship with call rate and reflectivity on their own, migrant height was not a significant predictor of the correlation between estimates from the 2 methods. It is possible that as average migrant height increases, more birds are flying above the range of acoustics recorders and above the range of radar, resulting in lower estimates from both methods. Future work analyzing typical flight heights of migrant species that pass through our study region, what determines flight heights, and how calling rates vary at differing flight heights may be needed to explain the trends found here. The association between methods decreased with increasing cloud cover and moon illumination. This relationship could be explained by the disproportionate impact that cloud cover and moon illumination had on call rate in relation to reflectivity. Additionally, while not analyzed by our study, ground cover and reflectance and its interaction with cloud cover and moon illumination could contribute to variability in avian migrant behavior (whether this be calling or flight behavior). Accounting for these variables may improve the strength of the correlation in estimates from each method and may also help to explain some of the site-to-site variability in estimates from acoustic monitoring that we discovered.

Although we found strong correlations between estimates of migration activity derived from each method, the relationship was not perfect. One overarching factor that may contribute to this unexplained variation is the potential inter-species variability in both calling and flight behaviors. Little is known about calling behavior or flight behavior for individual species, or how these behaviors vary within a night, throughout a migration season, or in response to weather and other variables (Farnsworth 2005). Additionally, the proportion of each species represented within the migrant community varies from night to night. If the calling or flight behaviors of certain species increase correlation in measures of migration activity from radar and acoustics, then nights when these species represent a large proportion of the migrant community will have higher correlations than nights when the proportion of these species is smaller. An improved understanding of how call rate and flight behavior varies between species and across different temporal and weather factors is needed to further refine our ability to analyze avian migration using these methods.

Analyses Using Lagged Radar Data

Unlike previous studies, the areas sampled by weather radar and the areas sampled by acoustics did not directly overlap in our study. Nonetheless, we found generally positive relationships between estimates of migration activity from each method across nights, comparable to studies where the areal coverage of each method did overlap. However, the relationship between methods did increase for most comparisons when the estimate of migration activity derived from weather radar was lagged by 2 deciles (~2 hr) to account for the distance between our radar and acoustic sampling areas. This result suggests that our acoustic recorders were able to capture fine-scale variation in migration intensities. Future studies should examine to what extent correlations between acoustic and radar-based estimates of migration activity persist as the sampling areas for each method become more distant, and analyze how local topographic features and potential migration corridors may impact these correlations.

Conclusions and Future Directions

For researchers to understand what may be causing variability in migration estimates between weather radar and acoustics more fully, and to improve our understanding of the migration behavior of avian migrants in general, the study of individual migrants with onboard acoustic recording instrumentation, altimeters, and accelerometers, or the use of three-dimensional arrays, will be beneficial. An improved understanding of individual migration behavior, how it varies between species, and how it impacts estimates of migration activity will allow researchers to select the most appropriate data type. For instance, a current hurdle in predicting avian migration at local scales is the ability of researchers to extrapolate radar-based estimates of migration activity to areas outside of radar sampling ranges. If researchers were able to explain the relationship between radar and acoustic measures of migration activity more fully, then estimates of migration activity from acoustic recording sites located in the area of extrapolation could be used to assess the performance of models predicting migration in these areas.

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Supplementary material

Supplementary material is available at *Ornithological Applications* online.

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Ethics statement

This study did not involve the capture or handling of animals, and no behaviors or actions were altered by the methods used.

Author contributions

D.M.O., T.F.W, and M.J.D. conceived the idea of the study. M.J.D. and T.F.W. obtained funding. D.M.O. collected the radar and acoustic data. B.M.V.D. processed the acoustic data. K.G.H. aided in processing and analyzing radar data. B.M.V.D. and F.A. assisted in the statistical analyses that were implemented. D.M.O. analyzed the data and wrote the first draft of the manuscript. All authors provided input, reviewed, and edited the manuscript.

Data availability

Analyses reported in this article can be reproduced using the data and code provided by Osterhaus et al. (2024).

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